

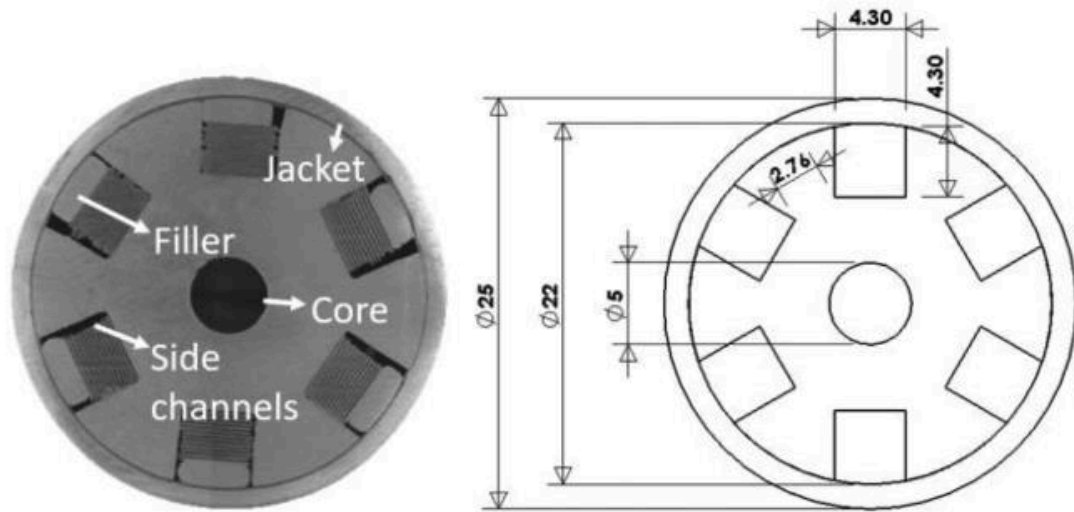
Electromagnetic modelling of superconducting cables

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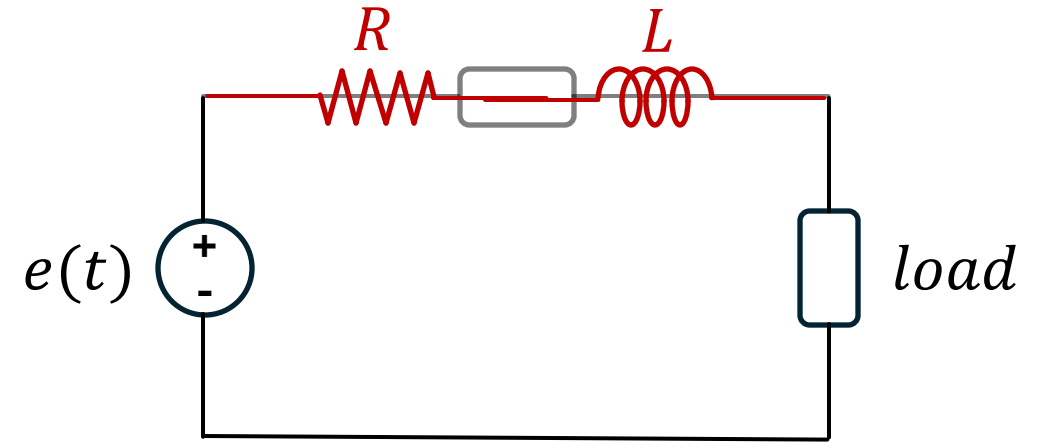
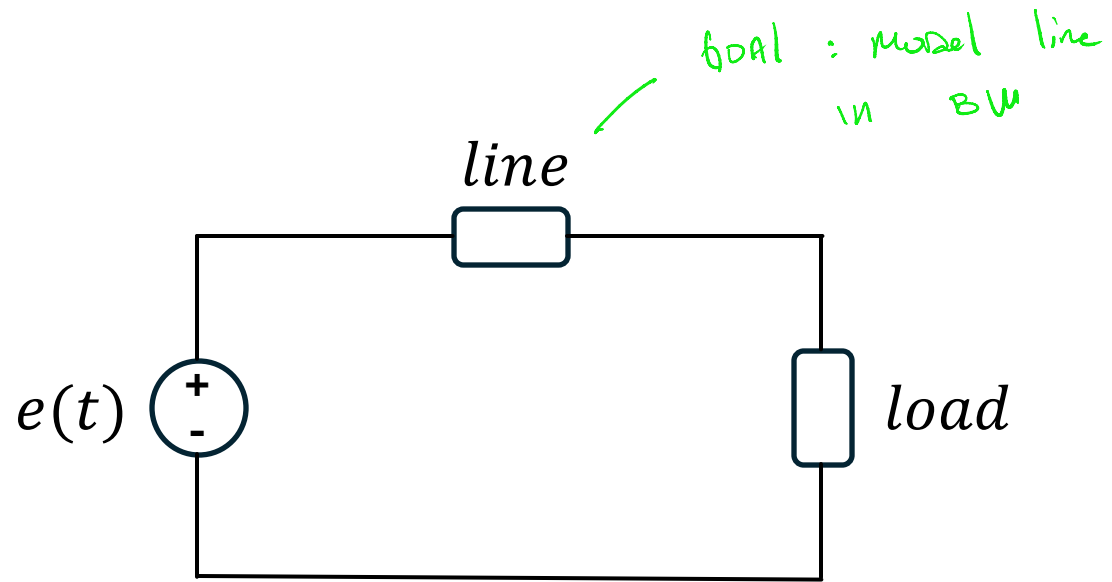
Aim of this lecture



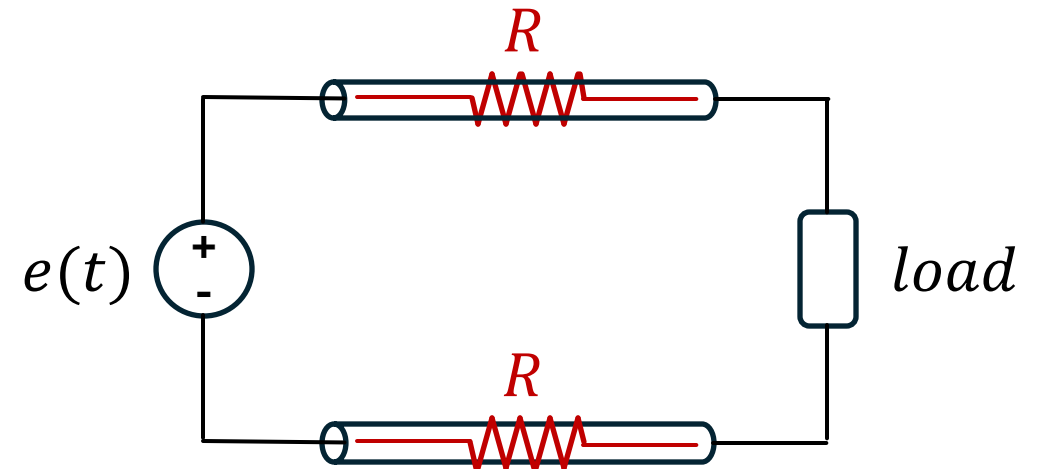
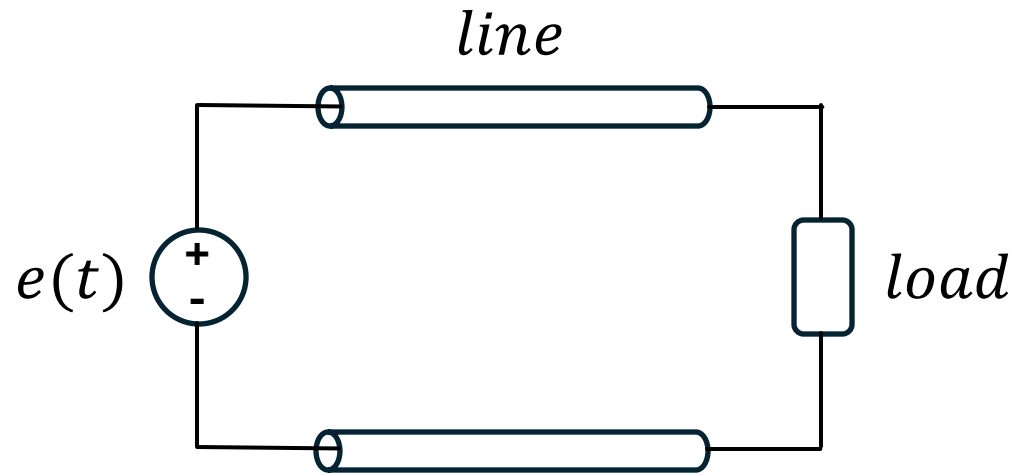
ENEA HTS CICC

- aluminum core
- six twisted slots with 0.5 m twist pitch
- each slot is equipped with 20 nonsoldered REBCO tapes having a width of 4 mm
- tapes are held in position using an aluminum filler,
- The cable is then inserted in an aluminum pipe (1.5-mm thick).

From circuit theory...



Cable resistance



$$R = \rho \frac{l}{S}$$

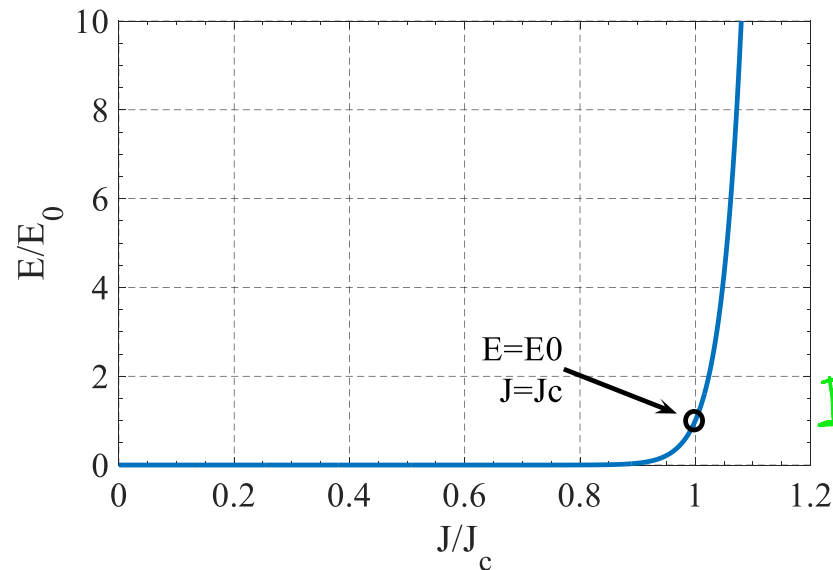
resistivity ρ *length* l *cross-section* S
no in s.c. condition

Nonlinear resistivity

$$\vec{J} = \sigma \vec{E} \Rightarrow \vec{E} = \rho \vec{J}$$

Macroscopic modelling: power law $E(J)$

$$E = E_0 \left(\frac{J}{J_c} \right)^n \Rightarrow \rho = \frac{E}{J} = \frac{E_0}{J_c} \left(\frac{J}{J_c} \right)^{n-1}$$



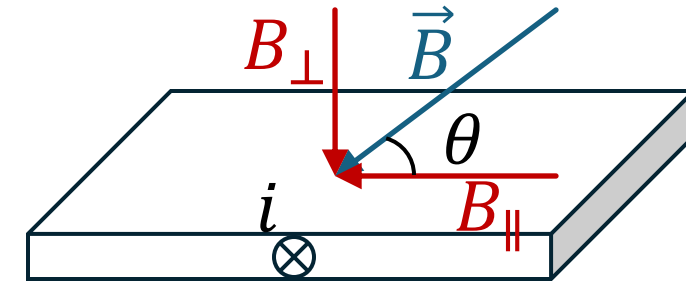
I very steep - 75% of J increase the resistivity

Definition of J_c : dependance on magnetic flux density

parameters are specific for one tape | interpolation of extrapolated data

reuse J_c

Model	Legend	Simplified Description	Material Law $J_c =$	Microstructure Parameters
Critical-State (CS)-Like model [23,39]	C_M	$J = \pm J_c$	J_{c0}	$J_{c0} = 2.85 \times 10^{10} \text{ A/m}^2$ [42]
Kim's Model [40]	K_{M1}	$J_c(\mathbf{B})$	$\frac{J_{c0}}{\left(1 + \frac{ \mathbf{B} }{B_0}\right)^\alpha}$	$B_0 = 240 \text{ mT}$ $\alpha = 1.5$ [22,42]
Kim-Like model [47]	K_{M2}	$J_c(B_{\parallel}, B_{\perp})$	$\frac{J_{c0}}{\left(1 + \frac{\sqrt{k^2 B_{\parallel} ^2 + B_{\perp} ^2}}{B_0}\right)^\alpha}$	$B_0 = 42.65 \text{ mT}$ $\alpha = 0.7$ $k = 0.29515$ [33,42,47] mitigates $B_{\parallel}$
Magneto-Angular Anisotropy Model [20]	R_M	$J_c(\mathbf{B}, \theta)$ Almost the same equation	$\frac{J_{c0}}{\left(1 + \epsilon_{\theta} \left(\frac{ \mathbf{B} }{B_0}\right)^\beta\right)^\alpha}$	$\epsilon_{\theta} = \sqrt{\gamma^{-1} \sin^2(\theta) + \cos^2(\theta)}$ $B_0 = 240 \text{ mT}, \alpha = 1.5$ $\beta = 1, \gamma = 5.02$ [20,22,42]



$$-90^\circ \leq \theta \leq 90^\circ$$

$$|\mathbf{B}| = \sqrt{B_{\perp}^2 + B_{\parallel}^2} =$$

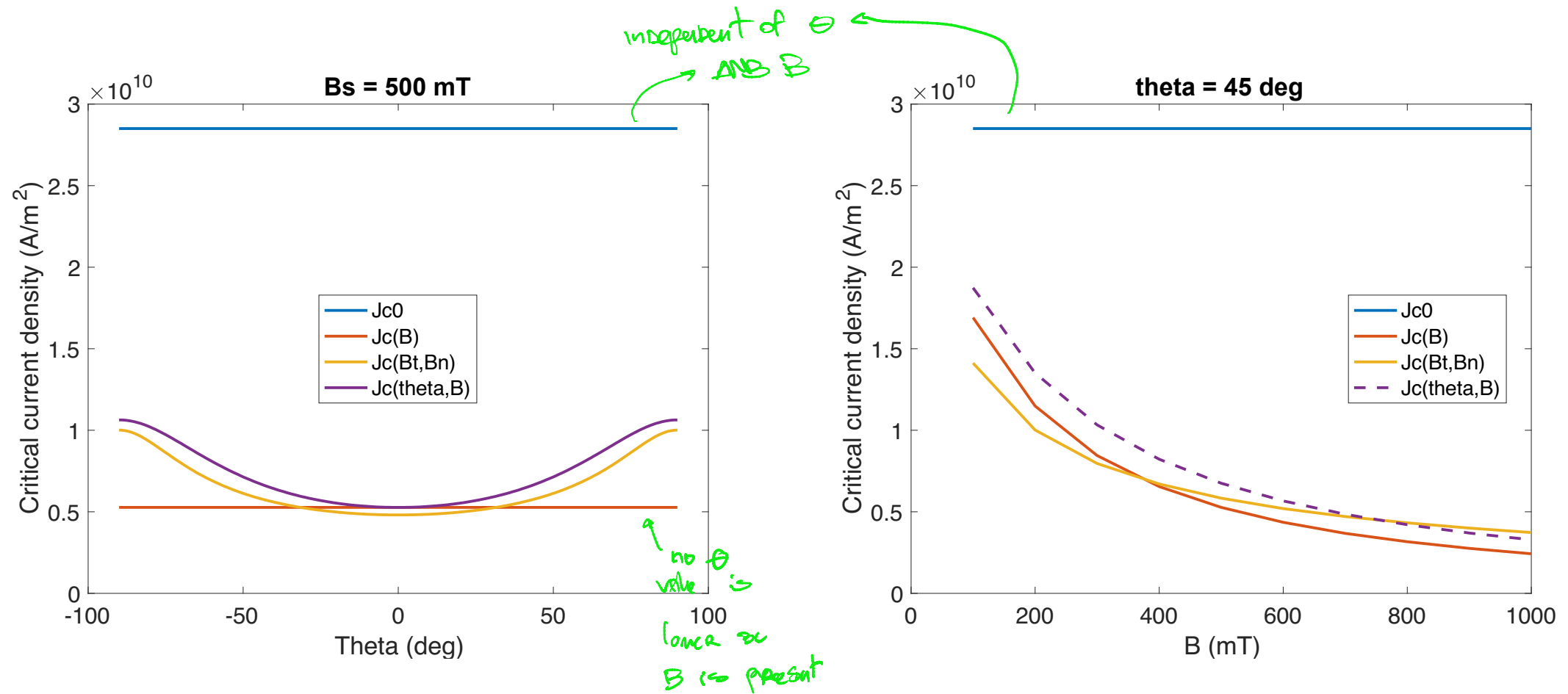
$$B_{\parallel} = B \cos \theta \rightarrow \cos$$

$$B_{\perp} = B \sin \theta \rightarrow \sin$$

$$= \sqrt{B^2 \cos^2 \theta + B^2 \sin^2 \theta}$$

$$\epsilon_0 = \sqrt{\gamma^{-1} \left(\frac{B_{\perp}}{B} \right)^2 + \left(\frac{B_{\parallel}}{B} \right)^2} = \frac{1}{B} \sqrt{\gamma^{-1} B_{\perp}^2 + B_{\parallel}^2} \rightarrow \left(1 + \epsilon_0 \frac{B}{B_0} \right)^2 = \left(1 + \frac{\sqrt{\gamma^{-1} B_{\perp}^2 + B_{\parallel}^2}}{B_0} \right)^2$$

Comparison



Definition of J_c : dependance on temperature and strain

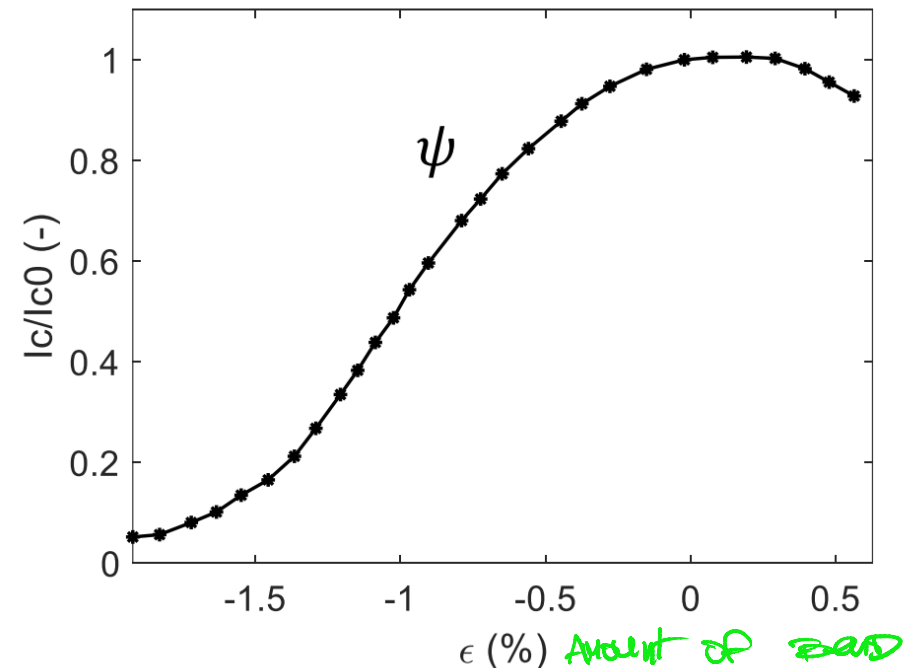
Scaling of J_c with temperature

$$J_c = J_{c0} \left(1 - \frac{T}{T_c} \right)^s$$

with $1.9 \leq s \leq 2.6$

Q. Wang *et al.*, "Temperature dependence of transport critical current density for REBCO superconducting tapes deposited by MOCVD," in *Superconductor Science and Technology*, Volume 37, Number 1, pp. 1-12, 2024, Art no. 015006, doi: 10.1088/1361-6668/ad0db5.

Strain dependance



D.C. van der Laan *et al.*, "Status of CORC® cables and wires for use in high-field magnets and power systems a decade after their introduction," in *Superconductor Science and Technology*, Volume 32, Number 3, pp. 1-12, 2019, Art no. 033001, doi: 10.1088/1361-6668/aafc82.

even in after some time I made it strength addition
it will have decrease of SC.

Self inductance

- Magnetic flux density

$$B \propto i \Rightarrow \Phi \propto i$$

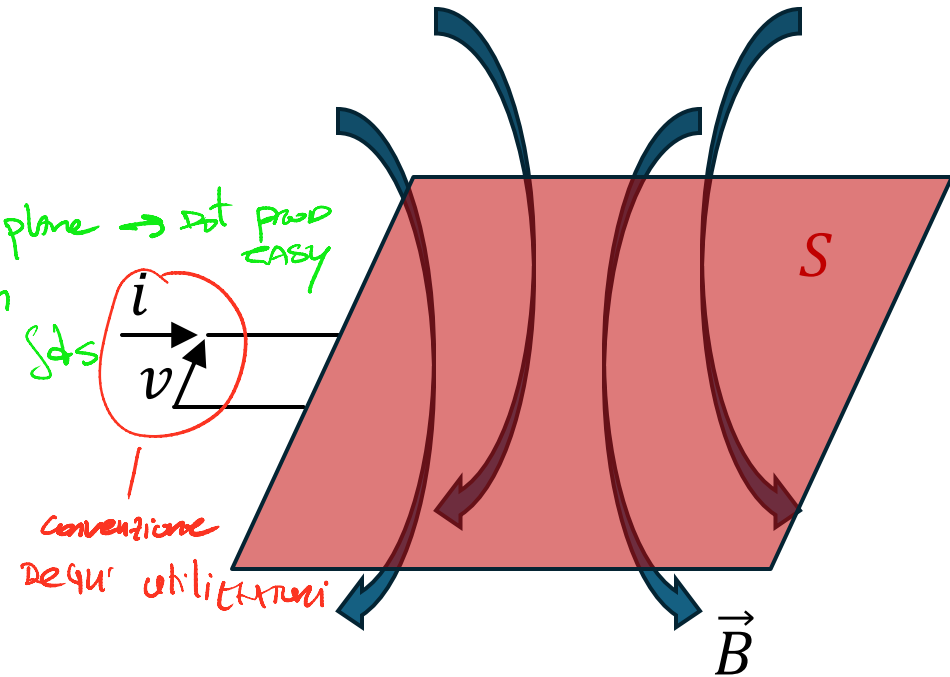
- Magnetic flux

$$\Phi = \int_S \vec{B} \cdot d\vec{S} = Li$$

2 ASSUMP

$B \perp$ to plane $\rightarrow$ dot prod EASY
 B uniform

B out of Sds



conventions
requ' utilization

- Faraday's law

$$v = \frac{d\Phi}{dt} = L \frac{di}{dt}$$

Voltage

bc of $\Phi \propto i$

inductance angle, length of the surface (geometry property) of the coil
but also introduces property of the material

$$J_c = \frac{J_{c0}}{\left(1 + \frac{BL}{B_0}\right)^2} \left(1 - \frac{T}{T_c}\right)^5 \psi$$

not linear?

$$J_c = \frac{J_{c0}}{\left(1 + \frac{B(J)}{B}\right)^\alpha} \left(1 + \frac{T(J)}{T_c}\right)^5 \psi$$

Electromagnetic time scale

term problem

x 100 time scale
of the e.m.
problem

Better for convergence of
high linear prob

↓
difficult to implement tho

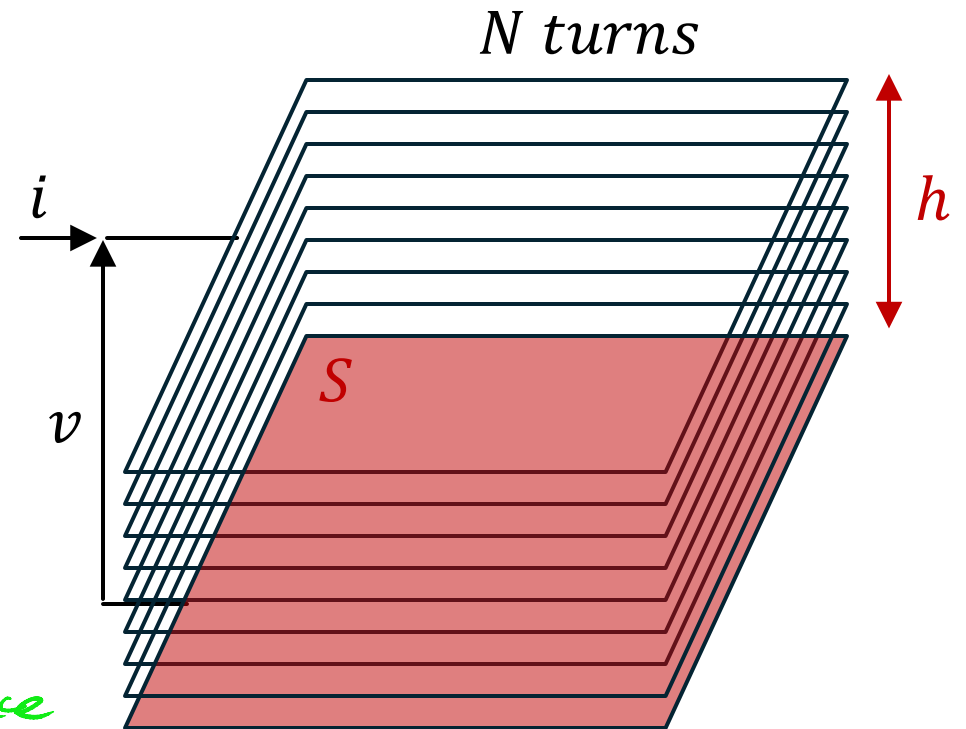
Self inductance

μ / many turns all coincident
 B N times stronger

- Inductance is associated to a loop
- The voltage (emf) is not associated to a specific position in the loop
- Measure of the “linked” flux

$$L = \mu N^2 \frac{S}{h}$$

medium — S — cross section
Amplification of B
and also electromotive force



Mutual inductance

— Flux induced by another coil
if $\frac{d\Phi}{dt}$ changes $\rightarrow$ electromotive force in the second coil

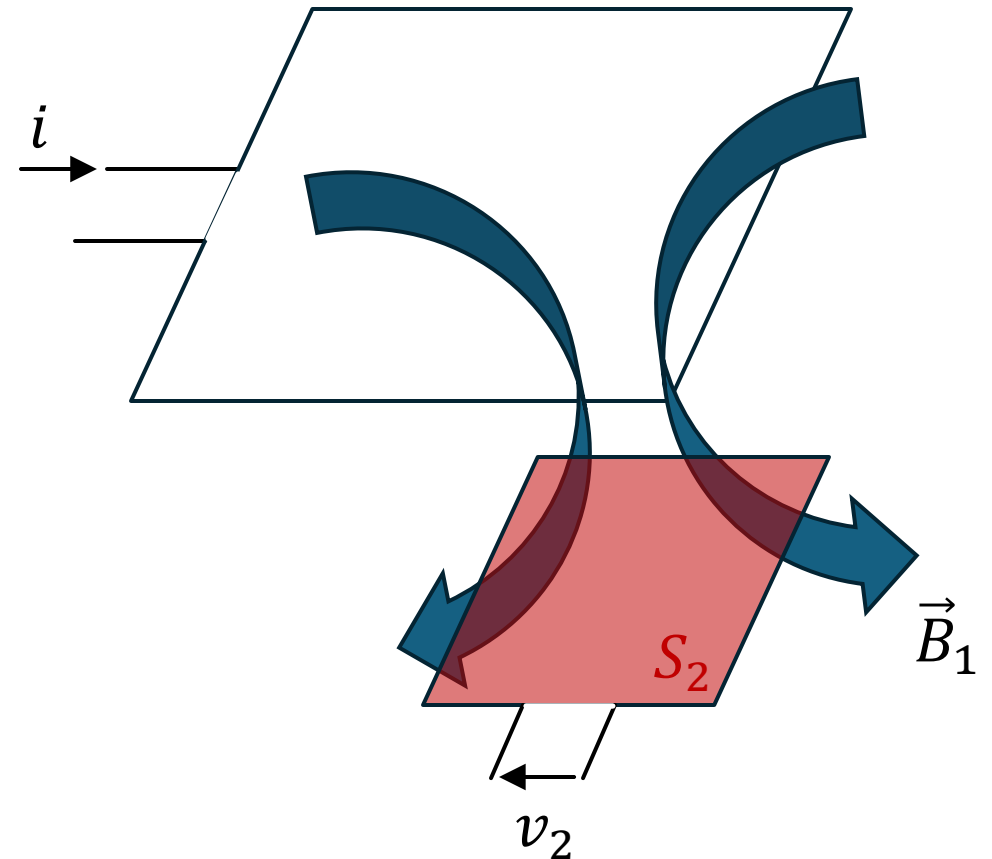
- A current i_1 flows in coil 1
- A magnetic flux density B_1 is generated by the current i_1
- Coil 2 links the magnetic flux density B_1

$$\Phi_{12} = \int_{S_2} \vec{B}_1 \cdot d\vec{S}$$

- A voltage v_2 is induced in coil 2

$$v_2 = \frac{d\Phi_{12}}{dt} = M_{12} \frac{di_1}{dt}$$

$\downarrow$
geometry coef. but also medium



Mutual inductance

- For N coupled loops

$$v_1 = \frac{d\Phi_1}{dt} = L_{11} \frac{di_1}{dt} + M_{12} \frac{di_2}{dt} + \dots + M_{1N} \frac{di_N}{dt}$$

$$v_N = \frac{d\Phi_N}{dt} = M_{N1} \frac{di_1}{dt} + M_{N2} \frac{di_2}{dt} + \dots + L_{NN} \frac{di_N}{dt}$$

With

$$L_{kk} = \left. \frac{\Phi_k}{i_k} \right|_{i_{j \neq k} = 0} \quad M_{kj} = \left. \frac{\Phi_k}{i_j} \right|_{i_{m \neq j} = 0}$$

Partial inductance

- However, we want to model a cable, not a loop!
- Let's re-work the inductance formula

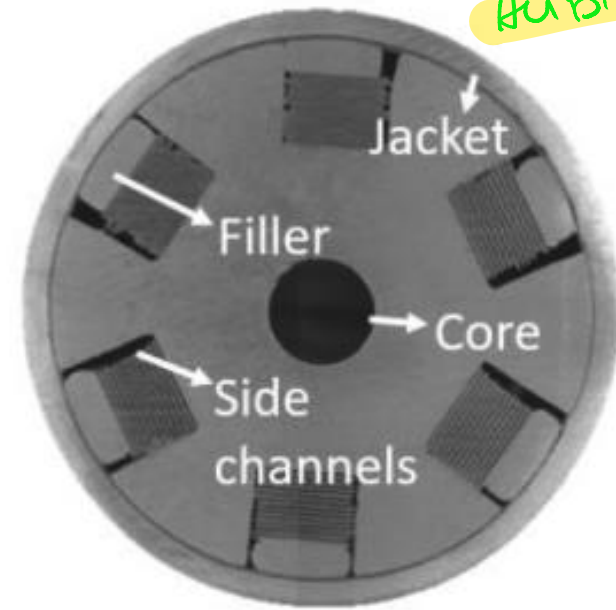
$$L = \frac{\Phi}{i} = \frac{1}{i} \int_S \vec{B} \cdot d\vec{S}$$

- The magnetic flux density is div-free $\rightarrow$ magnetic vector potential

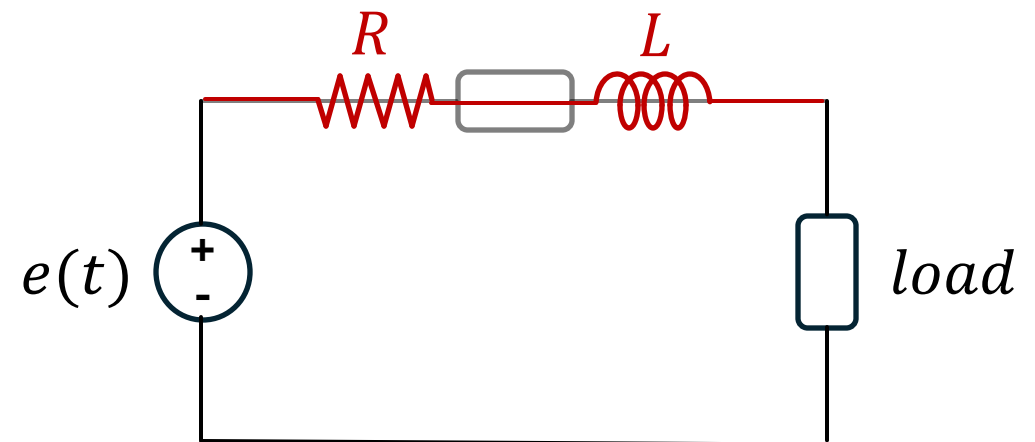
$$\nabla \cdot \vec{B} = 0 \Rightarrow \vec{B} = \nabla \times \vec{A}$$

divergence free

$\hookrightarrow \nabla \cdot \vec{B} = \nabla \cdot (\nabla \times \vec{A}) = 0$ divergence of a curl is always zero



Audio 1.15



$\nabla \times \vec{A}$ $\rightarrow$ MAGNETIC VECTOR POTENTIAL
 $\oint_C \vec{A} \cdot d\vec{s} = \iint_S (\nabla \times \vec{A}) \cdot d\vec{s}$
 $\oint_C \vec{A} \cdot d\vec{s}$ BOUNDARY OF THE AREA
 $\oint_C \vec{A} \cdot d\vec{s}$ BETTER THAN A DOUBLE INTEGRAL

Audio 1:02

inductance for segment
not just a loop

A diagram illustrating a surface (a parallelogram) with two vector fields. The surface is colored red. Two blue curved arrows represent the vector fields. The left arrow is labeled with a red 'A' and the word 'double' in red. The right arrow is labeled with a blue vector symbol $\vec{B}$.



A diagram showing a parallelogram with a red border. Inside the parallelogram, there is a blue circular arrow pointing clockwise. In the center of the circle is a vector symbol $\vec{A}$.



Partial inductance

$$\rightarrow L_1 = \frac{1}{i} \int_{l_1} \vec{B}_1 \cdot d\vec{l} \quad \text{there's no surface}$$

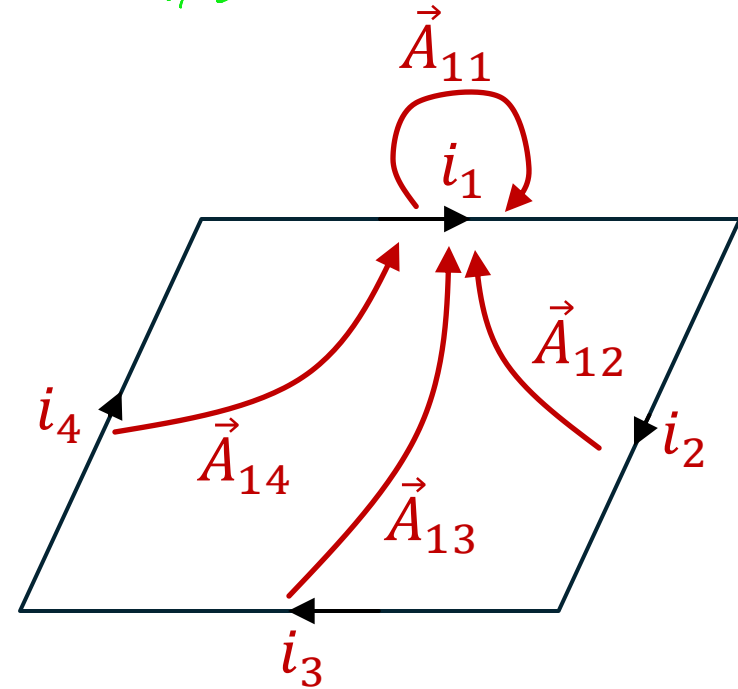
- Let's consider side 1

$$L_1 = \frac{1}{i} \int_{l_1} \vec{A}_1 \cdot d\vec{l}$$

- $\vec{A}_1$ has contributions from the currents of all sides

$$\vec{A}_1 = \vec{A}_{11} + \vec{A}_{12} + \vec{A}_{13} + \vec{A}_{14}$$

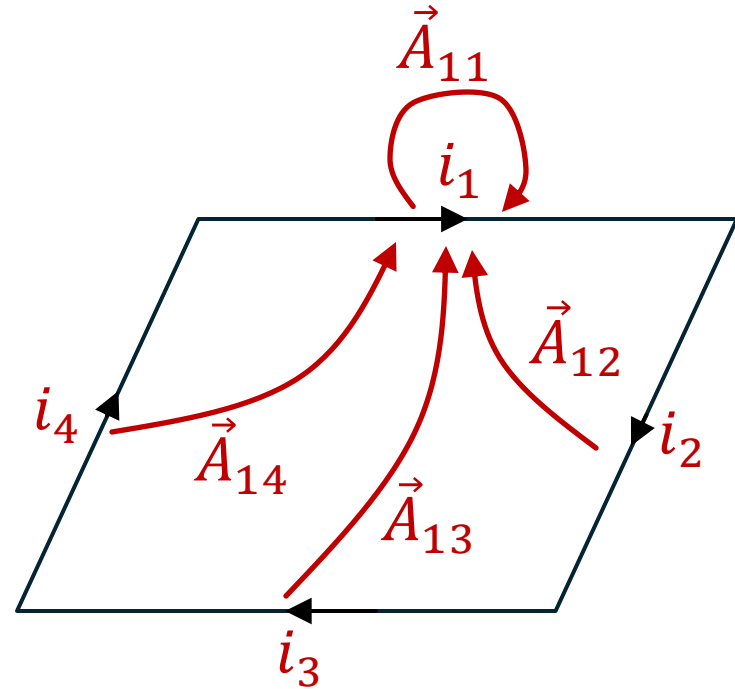
first
segment second
segment



Partial inductance

$$L_1 = \frac{1}{i} \left[\int_{l_1} \vec{A}_{11} \cdot d\vec{l} + \int_{l_1} \vec{A}_{12} \cdot d\vec{l} + \int_{l_1} \vec{A}_{13} \cdot d\vec{l} + \int_{l_1} \vec{A}_{14} \cdot d\vec{l} \right]$$
$$L_1 = L_{p11} + M_{p12} + M_{p13} + M_{p14}$$

and similarly for L_2, L_3, L_4



Partial inductance

The induced voltages on the 4 sides are

These coils are associated w/ segment not loops

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} \overset{\text{self}}{L_{p11}} & \overset{\text{mutual}}{M_{p12}} & M_{p13} & M_{p14} \\ M_{p21} & L_{p22} & M_{p23} & M_{p24} \\ M_{p31} & M_{p32} & L_{p33} & M_{p34} \\ M_{p41} & M_{p42} & M_{p43} & L_{p44} \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix}$$

And with $i_1 = i_2 = i_3 = i_4 = i$ and $v = v_1 + v_2 + v_3 + v_4$ we obtain the loop inductance

Partial inductances: calculation

- The magnetic vector potential due to a current in a segment is

$$\vec{A}_{kj} = \frac{\mu_0 i_j}{4\pi} \int_{l_j} \frac{1}{R_{kj}} \vec{dl}_j$$

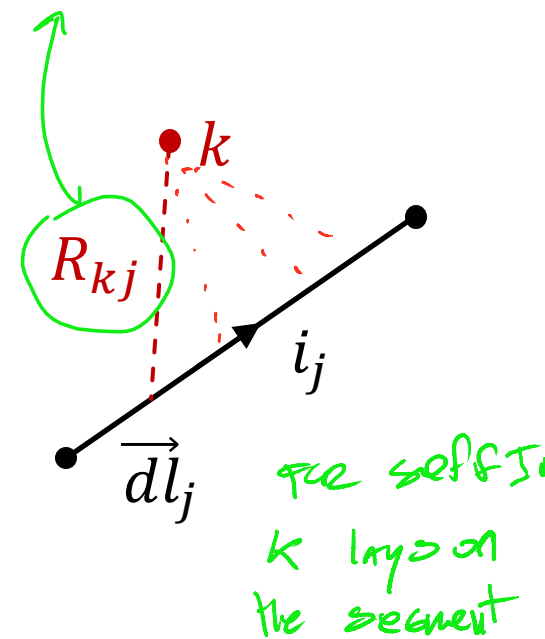
- Neumann formula

$$M_{p^{kj}} = \frac{1}{i_j} \int_{l_k} \left(\frac{\mu_0 i_j}{4\pi} \int_{l_j} \frac{1}{R_{kj}} \vec{dl}_j \right) \cdot \vec{dl}_k = \frac{\mu_0}{4\pi} \int_{l_k} \int_{l_j} \frac{\vec{dl}_j \cdot \vec{dl}_k}{R_{kj}} = M_{p^{jk}}$$

$$L_{pkk} = \frac{\mu_0}{4\pi} \int_{l_k} \int_{l_k} \frac{\vec{dl}_k \cdot \vec{dl}_k}{R_{kk}}$$

Singular!

DISTANCE SEGMENT FIELD
POINT K → IT'S A FUNC



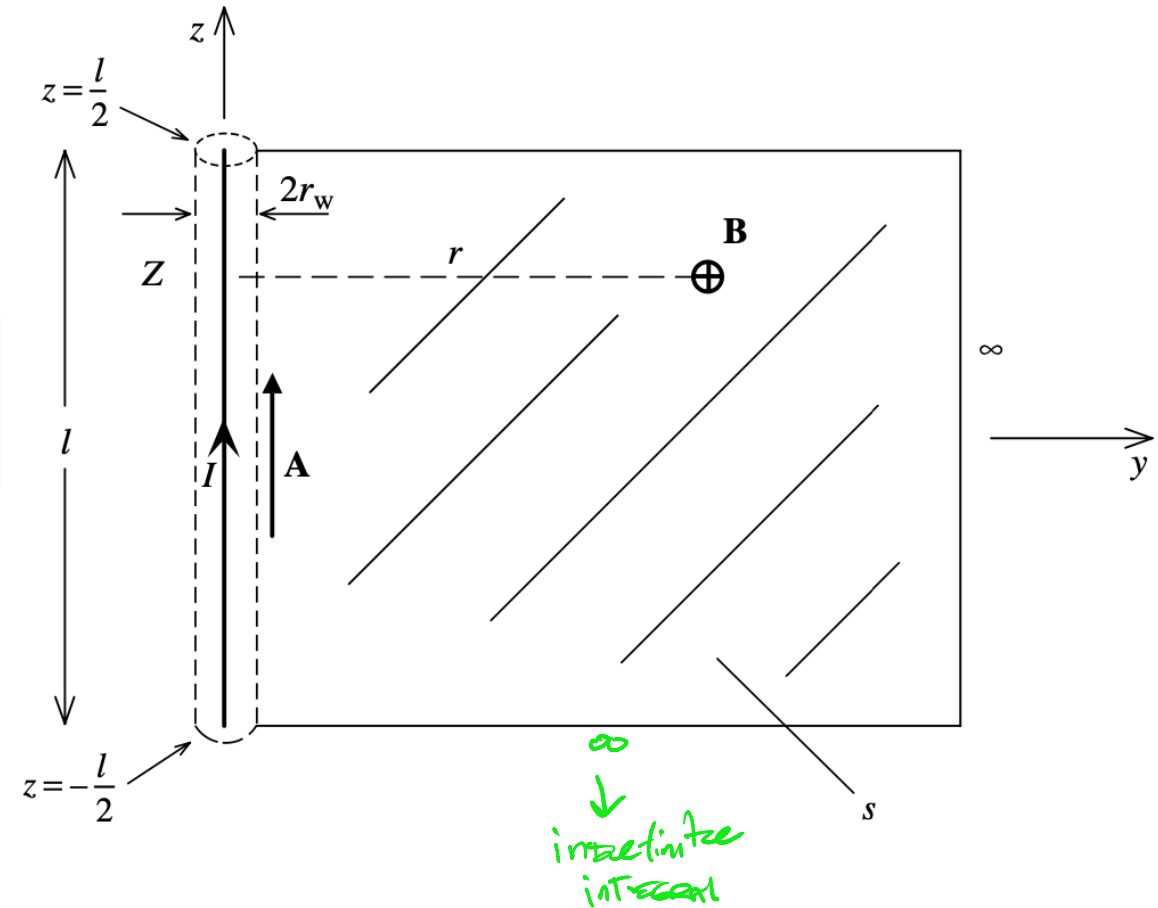
vuoto

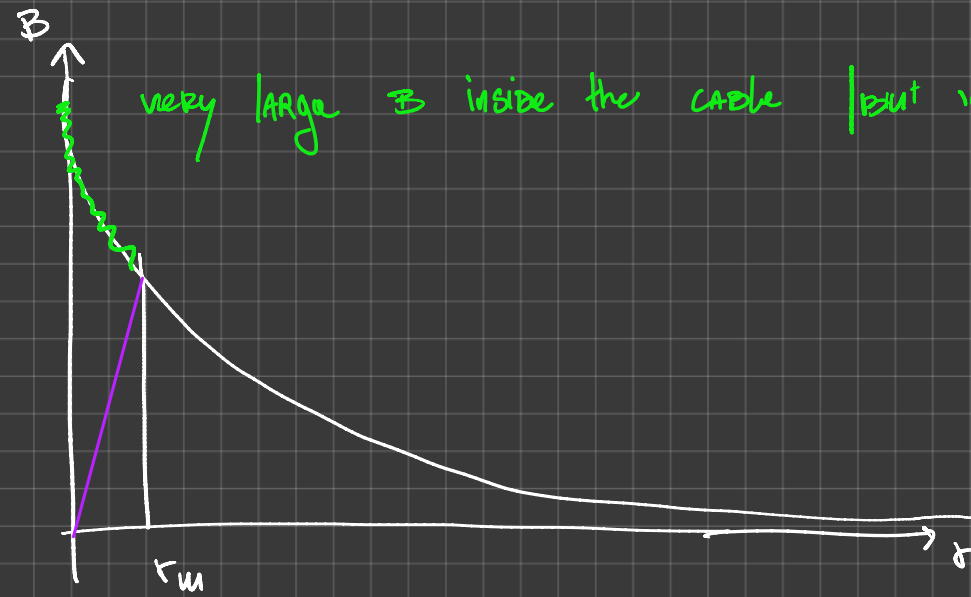
Geometry

ORDER
CHANGE
SYMMETRIC
MATRIX

Partial inductances: calculation

$$\begin{aligned}
 L_p &= \frac{\psi_\infty}{I} \\
 &= \frac{\mu_0}{2\pi} l \left[\ln \left(\frac{l}{r_w} + \sqrt{\left(\frac{l}{r_w} \right)^2 + 1} \right) - \sqrt{1 + \left(\frac{r_w}{l} \right)^2} + \frac{r_w}{l} \right] \\
 &= \frac{\mu_0}{2\pi} l \left[\sinh^{-1} \frac{l}{r_w} - \sqrt{1 + \left(\frac{r_w}{l} \right)^2} + \frac{r_w}{l} \right]
 \end{aligned}$$

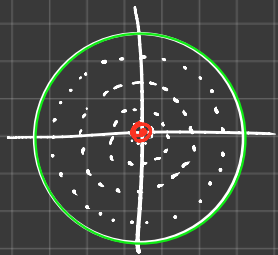




very large B inside the cable but inside the cable the current is lower



↑ HERE the current is zero ($r=0$), that's why actual B is going to zero



3.5

10 A

AND $1:52$

at the center there's no current

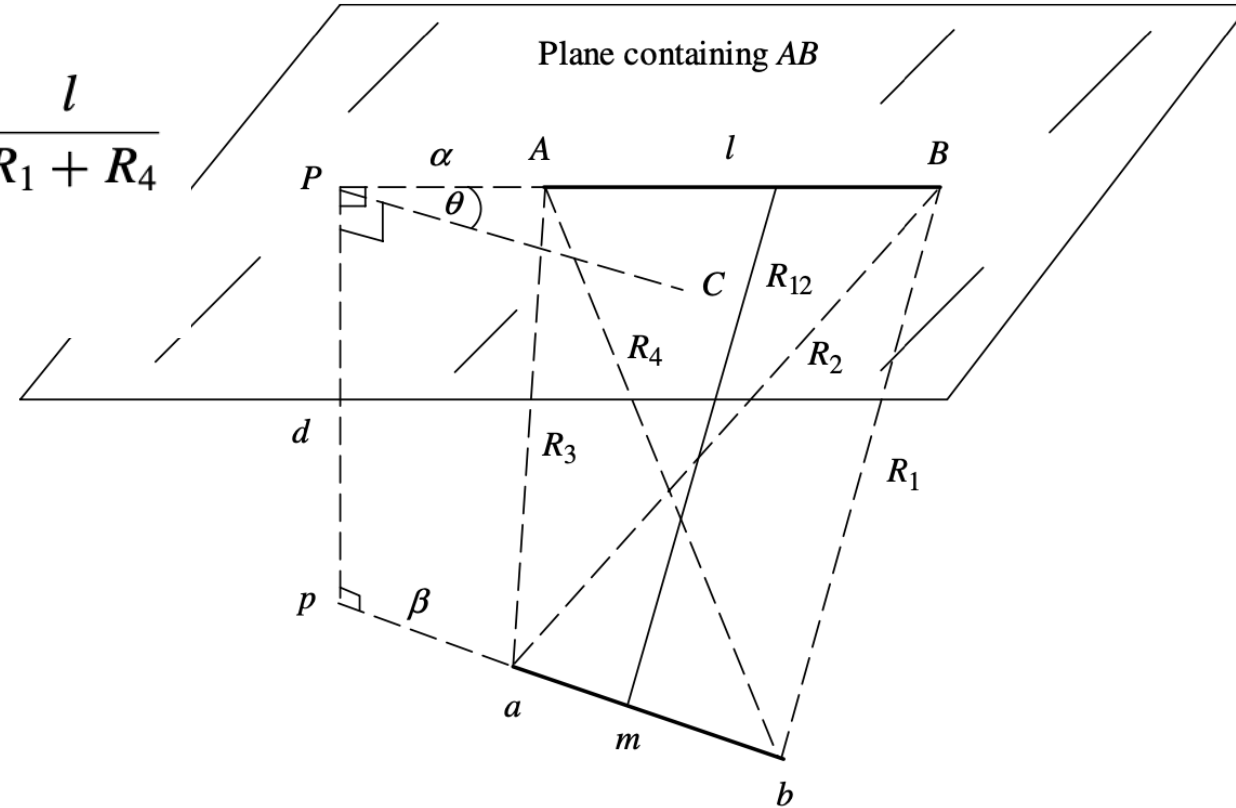
less linked current at the center

Partial inductances: calculation

they don't lay on the same plane
general use

$$M_p = \frac{\mu_0}{2\pi} \cos \theta \left[(\alpha + l) \tanh^{-1} \frac{m}{R_1 + R_2} + (\beta + m) \tanh^{-1} \frac{l}{R_1 + R_4} \right. \\ \left. - \alpha \tanh^{-1} \frac{m}{R_3 + R_4} - \beta \tanh^{-1} \frac{l}{R_2 + R_3} \right] - \frac{\mu_0}{4\pi} \frac{\Omega d}{\tan \theta}$$

$$\Omega = \tan^{-1} \frac{d^2 \cos \theta + (\alpha + l)(\beta + m) \sin^2 \theta}{dR_1 \sin \theta} \\ - \tan^{-1} \frac{d^2 \cos \theta + (\alpha + l)\beta \sin^2 \theta}{dR_2 \sin \theta} \\ + \tan^{-1} \frac{d^2 \cos \theta + \alpha\beta \sin^2 \theta}{dR_3 \sin \theta} \\ - \tan^{-1} \frac{d^2 \cos \theta + \alpha(\beta + m) \sin^2 \theta}{dR_4 \sin \theta}$$



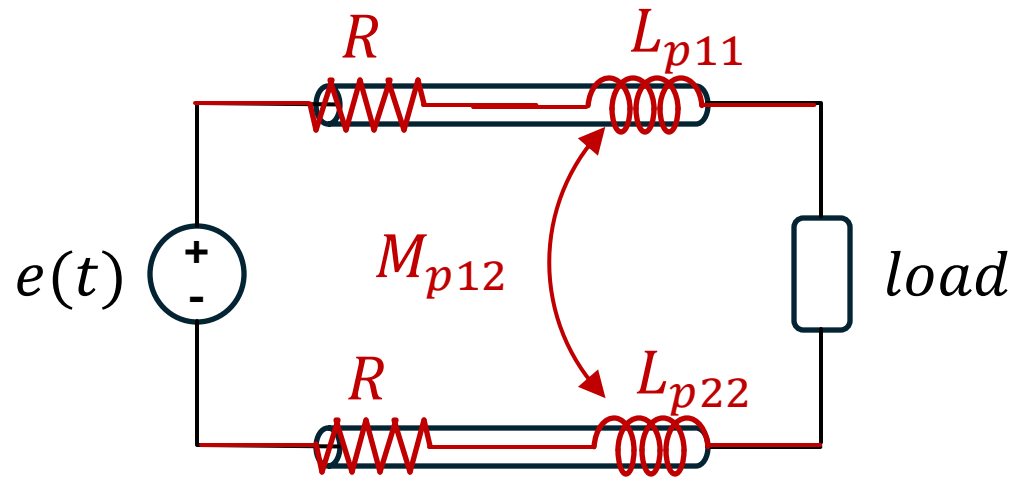
Partial inductances: calculation

Different analytic formulas for

- Parallel wires
- Parallel wires that are offset
- Aligned with offset wires
- Aligned touching
- Wires at an Angle to each other join at a common point
- Wires at an Angle to each other not touching
- Wires skewed and displaced

Or: numerical evaluation of the Neumann formula

Cable modelling



But: can we model the cable length with only one element?

- Match the geometry (joints, curves...)
- Catch physical phenomena (quench,...)
- Wavelength:

$$l < \frac{\lambda}{10} = \frac{1}{10} \frac{c}{f} = \frac{1}{10} c\tau$$

length of
the circuit

wave
length

if $\tau \approx 10 \mu s \rightarrow 6000$ km
BIGGER THAN THAT
it's not possible

Formulation from Maxwell Equations

- Mixed potential equation

$$\vec{E} = -\nabla\varphi - \frac{\partial\vec{A}}{\partial t}$$

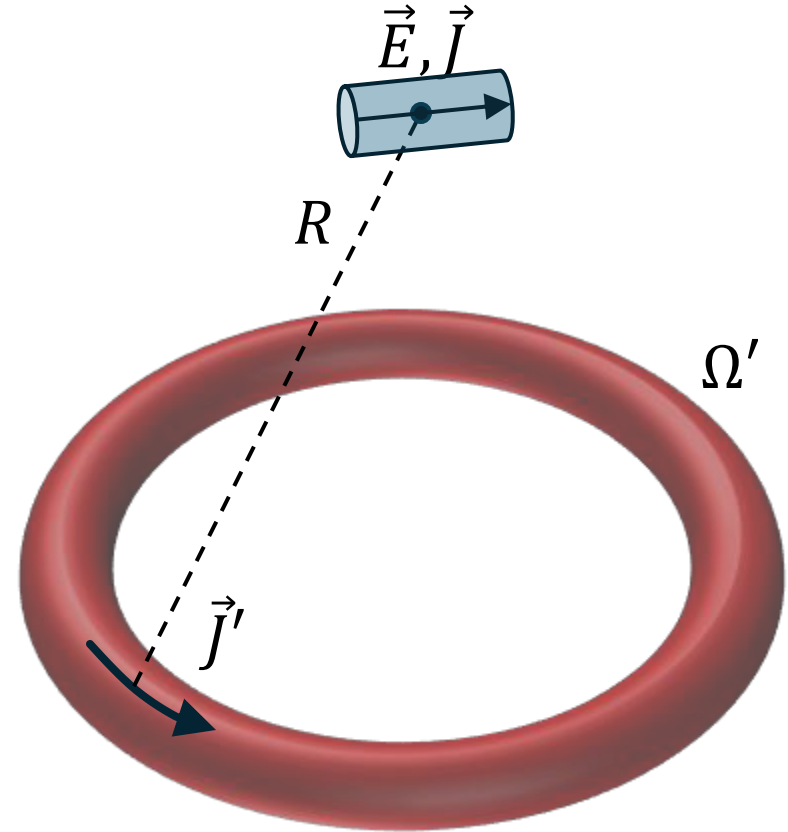
- Constitutive equation

$$\vec{E} = \rho\vec{J}$$

- Magnetic vector potential

$$\vec{A} = \frac{\mu_0}{4\pi} \int_{\Omega'} \frac{\vec{J}'}{R} d\Omega'$$

$$\rho\vec{J} + \frac{\partial}{\partial t} \frac{\mu_0}{4\pi} \int_{\Omega'} \frac{\vec{J}'}{R} d\Omega' + \nabla\varphi = 0$$



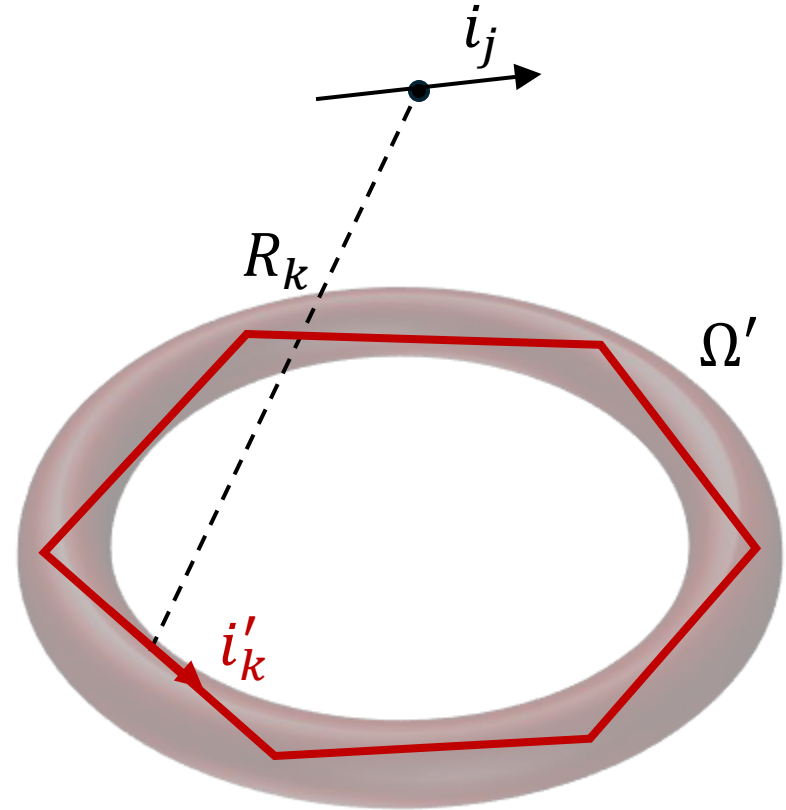
Formulation from Maxwell Equations

$$\rho \vec{J} + \frac{\partial}{\partial t} \frac{\mu_0}{4\pi} \int_{\Omega'} \frac{\vec{J}'}{R} d\Omega' + \nabla \varphi = 0$$

Hp: polygonal wire conductors

- $\vec{J}$ uniform in $S \rightarrow \vec{J} = \frac{i}{S} \vec{u}_l$
- $\vec{J}$ parallel to $\vec{dl}'$

$$\vec{A} = \sum_k \frac{\mu_0}{4\pi} \int_{l'_k} \frac{i'_k}{R_k} \vec{dl}'_k$$



Formulation from Maxwell Equations

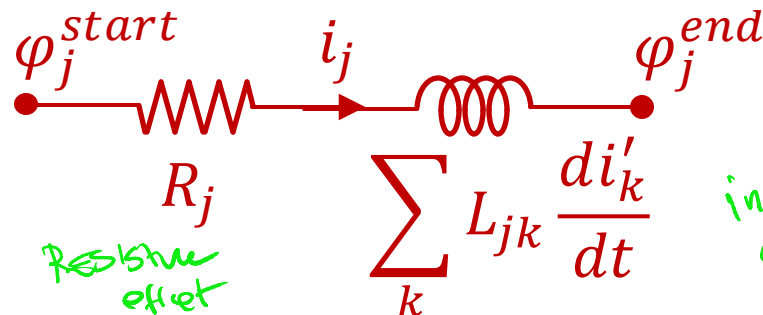
Integrating over the j th segment

$$\int_{l_j} \rho \frac{i_j}{S_j} \vec{u}_l \cdot d\vec{l}_j + \frac{\partial}{\partial t} \int_{l_j} \left[\sum_k \frac{\mu_0}{4\pi} \int_{l'_k} \frac{i'_k}{R_k} d\vec{l}'_k \right] \cdot d\vec{l}_j + \int_{l_j} \nabla \varphi \cdot d\vec{l}_j = 0$$

$$\rho \frac{l_j}{S_j} i_j + \sum_k \left[\frac{\mu_0}{4\pi} \int_{l_j} \int_{l'_k} \frac{i'_k}{R_k} d\vec{l}'_k \cdot d\vec{l}_j \right] \frac{di'_k}{dt} + (\varphi_j^{end} - \varphi_j^{start}) = 0$$

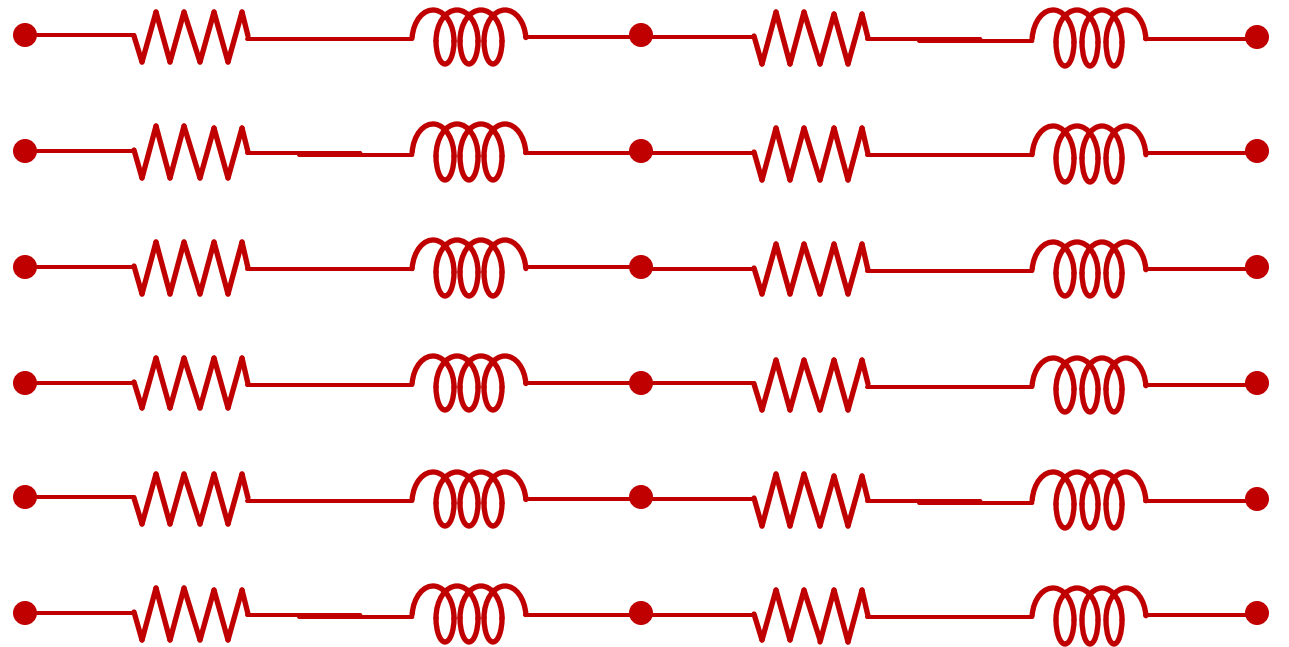
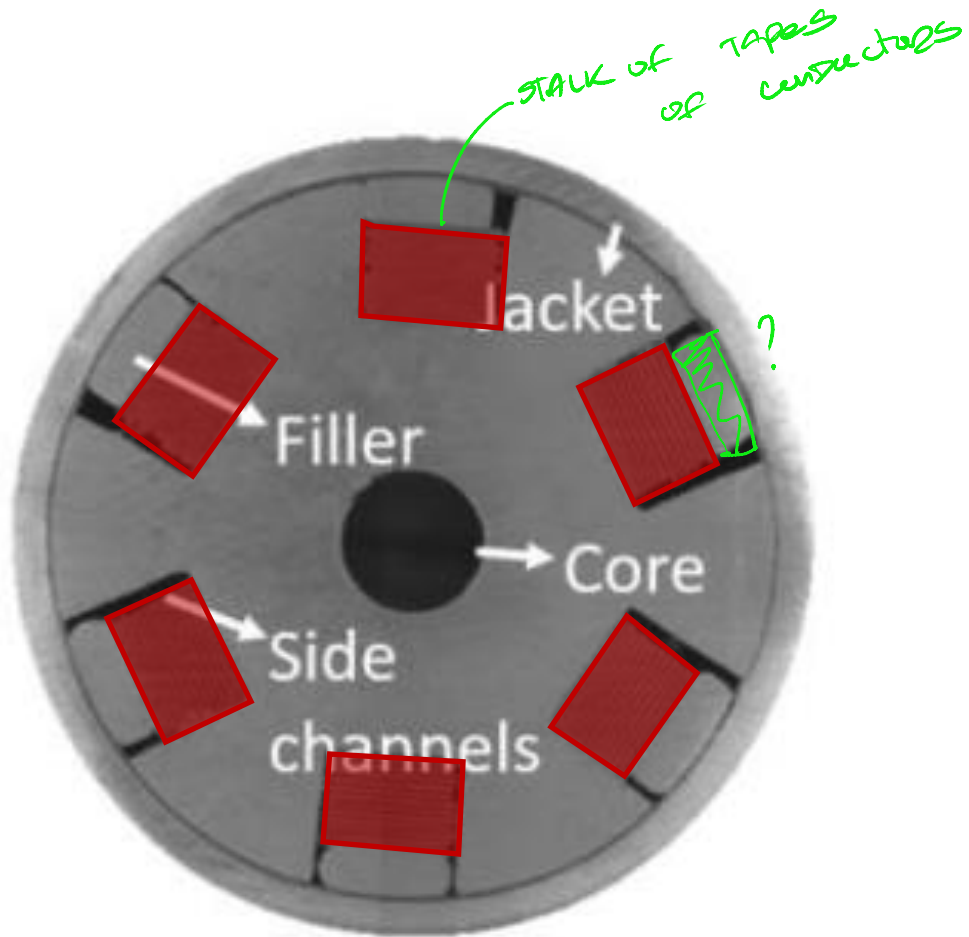
$$R_j i_j + \sum_k L_{jk} \frac{di'_k}{dt} = \varphi_j^{start} - \varphi_j^{end}$$

Voltage
Difference aka
potential difference
across the section

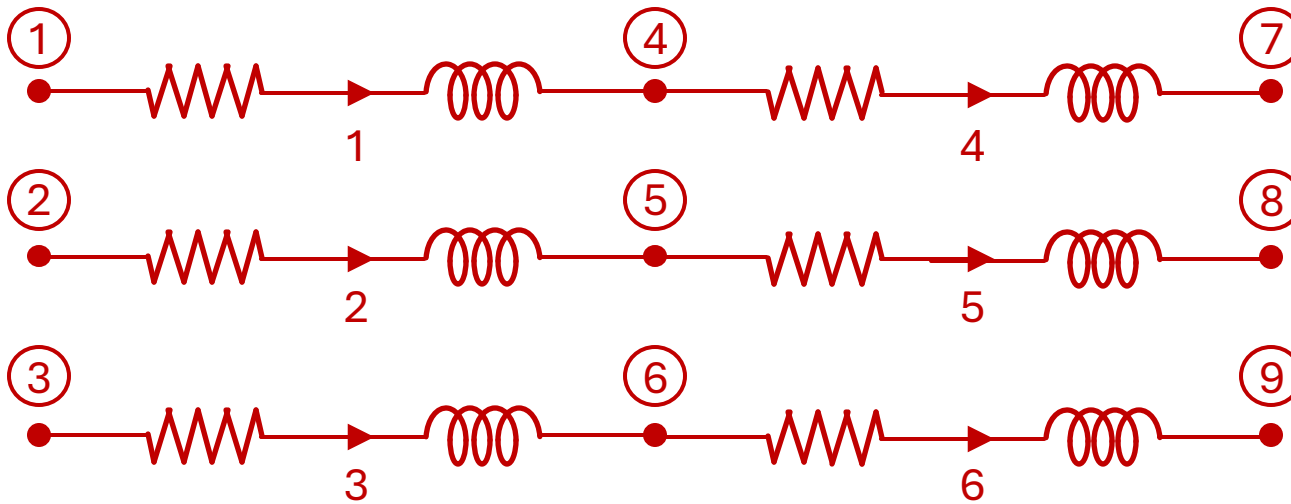


induction effect | COMBINATION
of all partial inductions

Electromagnetic modelling of SC cable



Incidence matrix



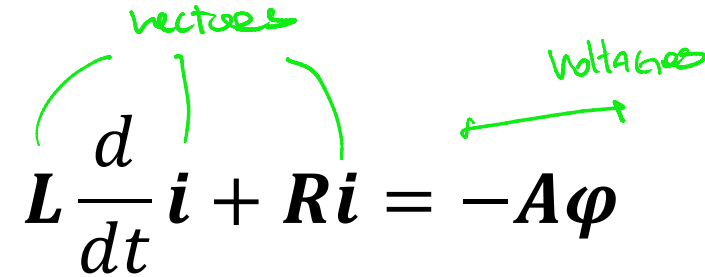
Nodes
 $N_N = 9$
 $N_E = 6$
Edges or segments

Edge-to-node incidence matrix

$$A = \begin{matrix} & \begin{matrix} \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} & \textcircled{5} & \textcircled{6} & \textcircled{7} & \textcircled{8} & \textcircled{9} \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} & \begin{bmatrix} -1 & & & 1 & & & & & \\ & -1 & & & 1 & & & & \\ & & -1 & & & 1 & & & \\ & & & -1 & & & 1 & & \\ & & & & -1 & & & 1 & \\ & & & & & -1 & & & 1 \end{bmatrix} \end{matrix}$$

$$-A\varphi = v$$

Equations in matrix form

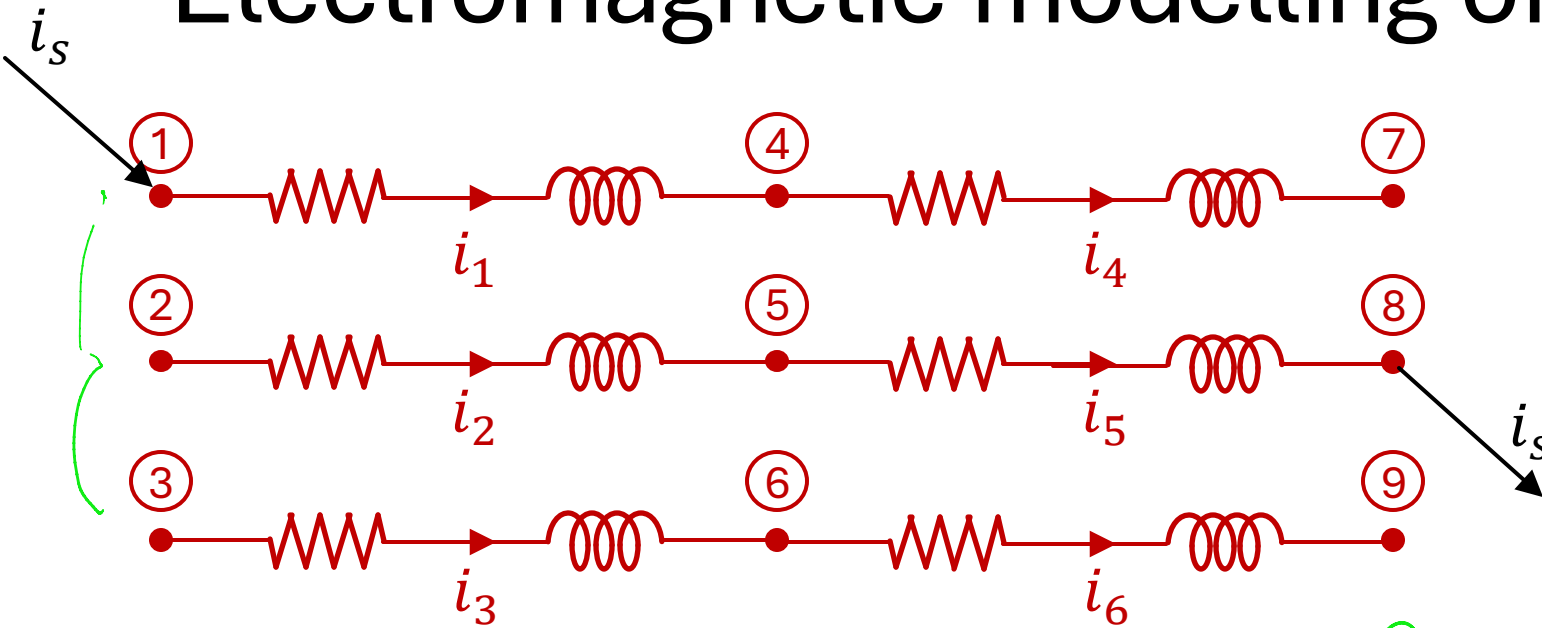


The diagram shows the matrix equation $L \frac{d}{dt} \mathbf{i} + R \mathbf{i} = -A \boldsymbol{\varphi}$. Handwritten green annotations include: a bracket above $\mathbf{i}$ labeled "vectors", a bracket above $\mathbf{i}$ and R labeled "vectors", and an arrow pointing from $\boldsymbol{\varphi}$ to the right labeled "voltages".

$$L \frac{d}{dt} \mathbf{i} + R \mathbf{i} = -A \boldsymbol{\varphi}$$

- R diagonal $N_E \times N_E$ resistance matrix
- L full $N_E \times N_E$ partial inductance matrix
- A edge-to-node $N_E \times N_N$ incidence matrix
- $\mathbf{i}$ vector of $N_E \times 1$ edge currents
- $\boldsymbol{\varphi}$ vector of $N_N \times 1$ nodal potentials

Electromagnetic modelling of SC cable



$$N_N = 9$$

$$N_E = 6$$

Kirchoff

KCL

$$A^T \mathbf{i} = \mathbf{i}_s$$

every node
for beginning and end

$$A = \begin{matrix} & \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} & \textcircled{5} & \textcircled{6} & \textcircled{7} & \textcircled{8} & \textcircled{9} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} & \begin{bmatrix} -1 & 1 & & & & & & & \\ & -1 & & & & & & & \\ & & -1 & & & & & & \\ & & & -1 & & & & & \\ & & & & -1 & & & & \\ & & & & & -1 & & & \end{bmatrix} \end{matrix}$$

20/03/2026

Audio

(Quasi) final equations

$$\begin{bmatrix} \mathbf{L} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \frac{d}{dt} \begin{bmatrix} \mathbf{i} \\ \boldsymbol{\varphi} \end{bmatrix} + \begin{bmatrix} \mathbf{R} & \mathbf{A} \\ -\mathbf{A}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{i} \\ \boldsymbol{\varphi} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{i}_s \end{bmatrix}$$

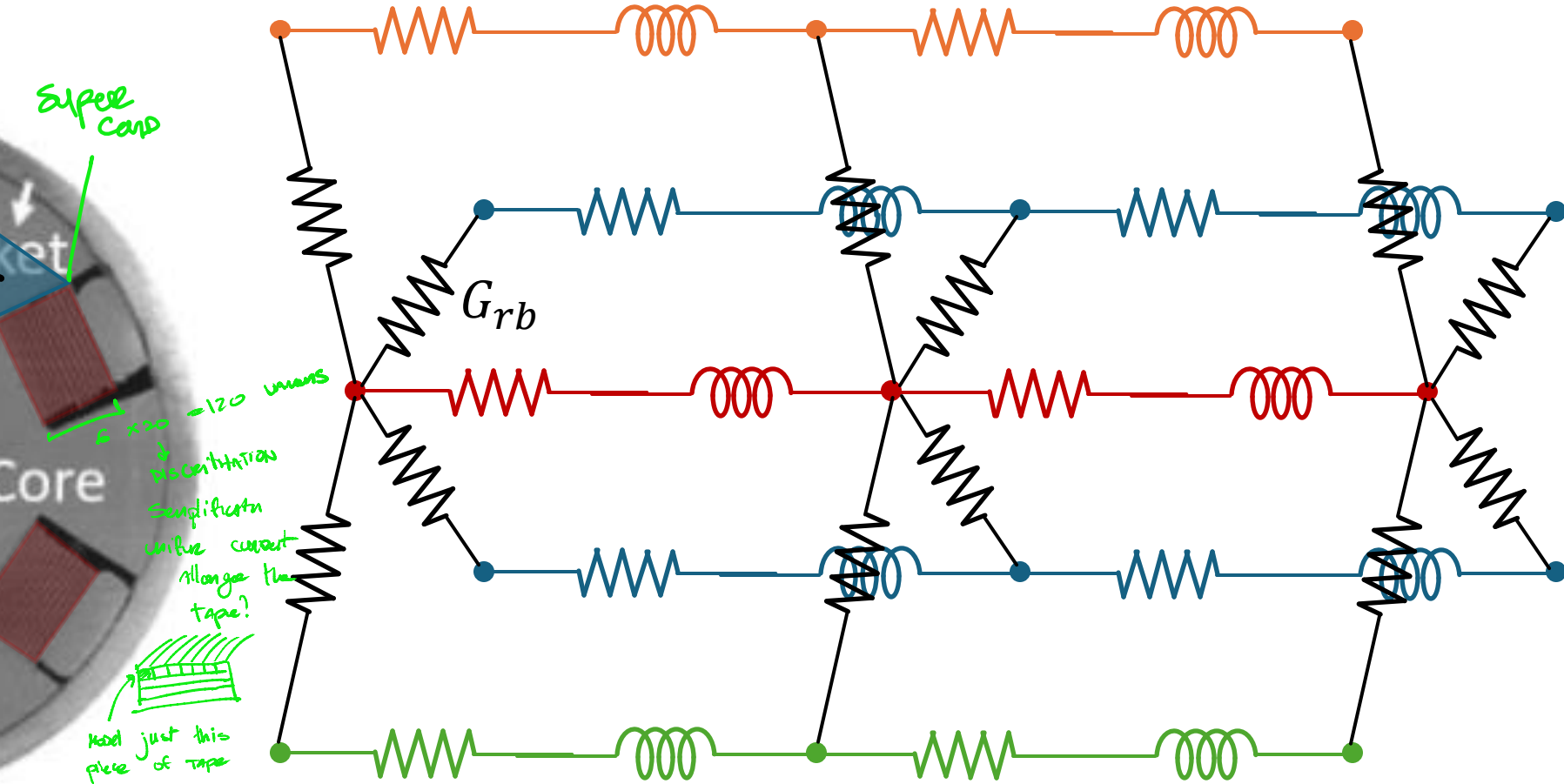
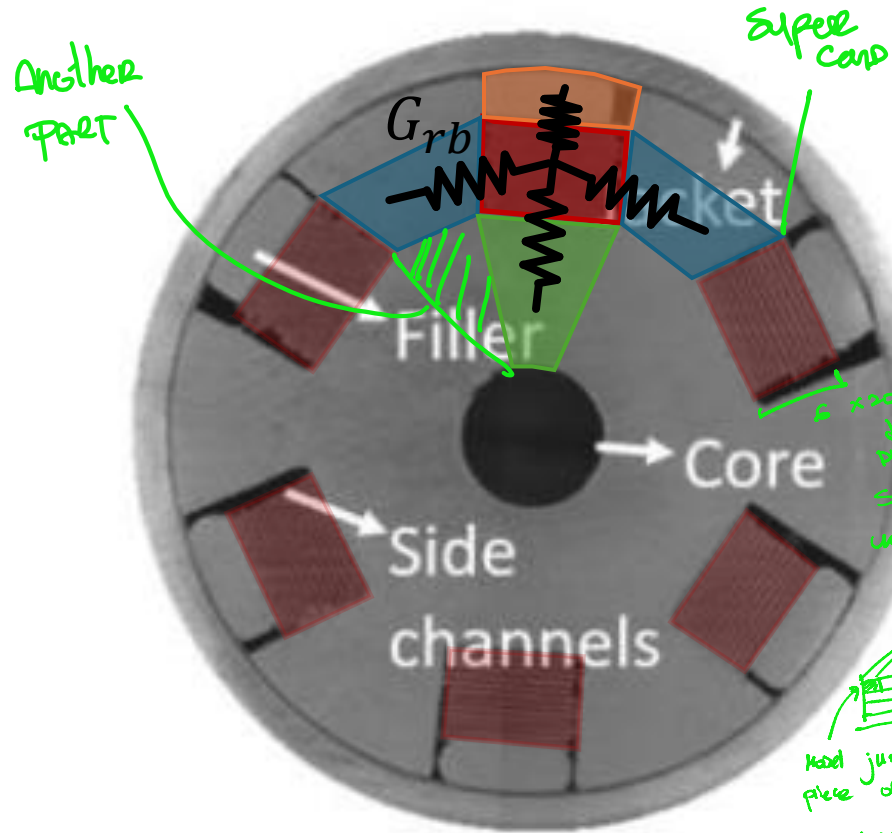
induct matrix

current from previous slide

$$\mathbf{M}\dot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{b}$$

- DAE equation: $\mathbf{M}$ is singular!
- Nonlinear: $\mathbf{R} = \mathbf{R}(\mathbf{i}, \mathbf{B}, \mathbf{T})$ environment too
- Dense: $\mathbf{L}$ is full (memory $O(N_E^2)$, solution $O(N_E^3)$)
- In AC: $(\mathbf{K} + j\omega\mathbf{M})\underline{\mathbf{x}} = \underline{\mathbf{b}}$

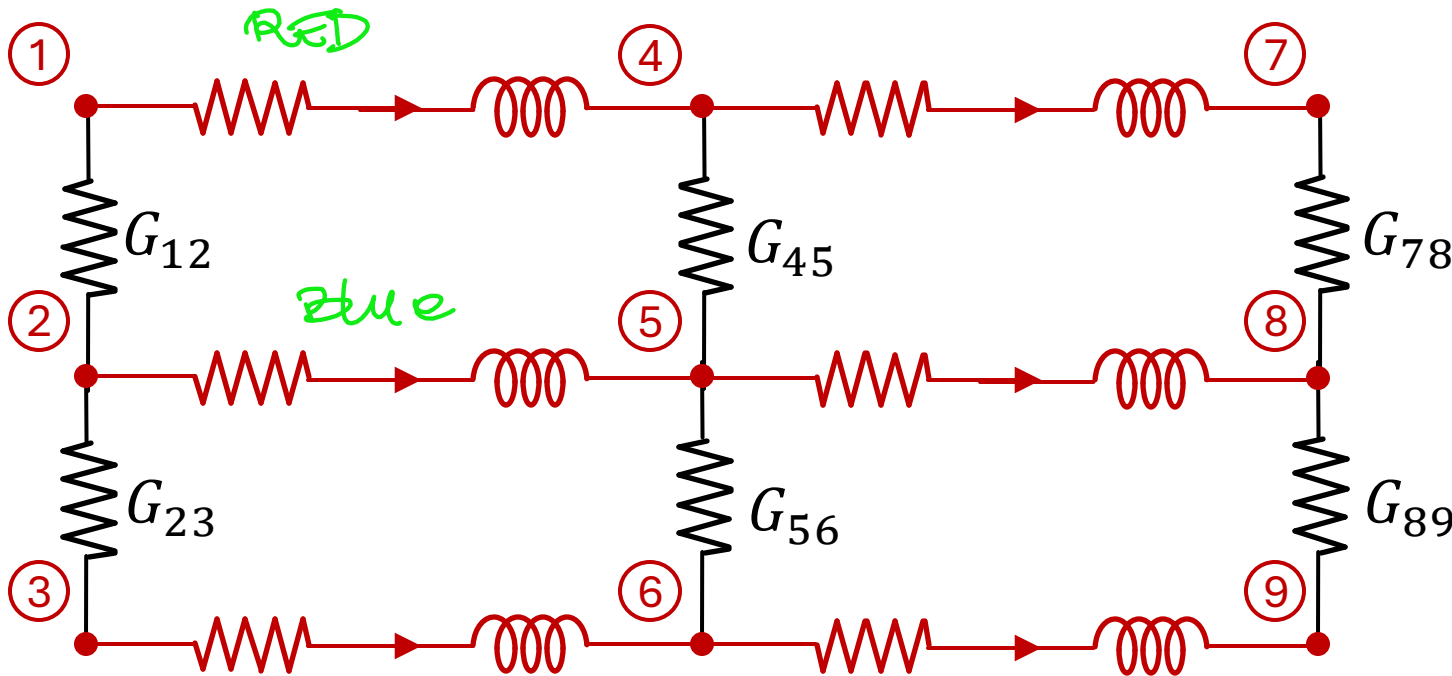
Transverse conductances



very complicated as L is full
 such n of unknown $\rightarrow n^2$ computational time $-n^2$ memory capability

Transverse conductances

simplification of previous slide



- New (black) edge-to-node incidence matrix $\mathbf{B}$
- Diagonal matrix $\mathbf{G}_d$ with conductances G_{ij}
- Voltage along black edges
 $\mathbf{v}_G = -\mathbf{B}\boldsymbol{\varphi}$
- Current through black edges
 $\mathbf{i}_G = \mathbf{G}_d \mathbf{v}_G = -\mathbf{G}_d \mathbf{B}\boldsymbol{\varphi}$
- KCL for black edges
 $\mathbf{B}^T \mathbf{i}_G = -\mathbf{B}^T \mathbf{G}_d \mathbf{B}\boldsymbol{\varphi} = -\mathbf{G}\boldsymbol{\varphi} = 0$

sparse diagonal sparse

Final equations

- KCL including all edge currents

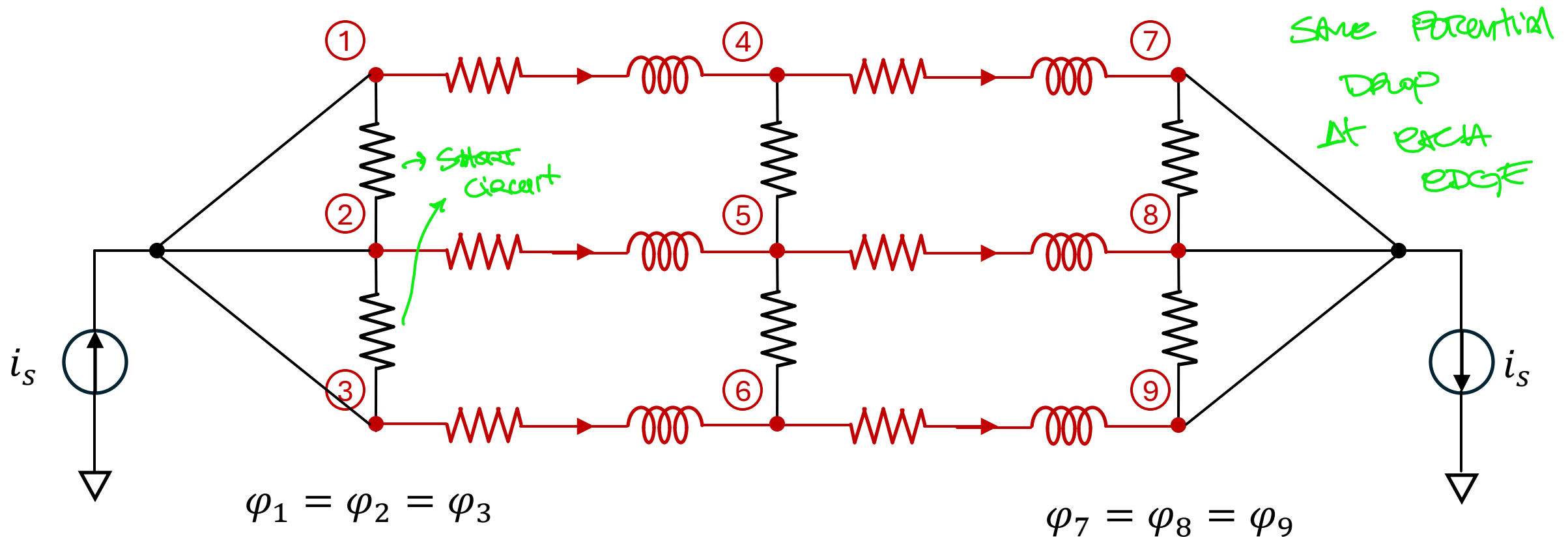
$$\mathbf{G}\boldsymbol{\varphi} - \mathbf{A}^T \mathbf{i} = \mathbf{i}_s$$

- Hence

$$\begin{bmatrix} \mathbf{L} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \frac{d}{dt} \begin{bmatrix} \mathbf{i} \\ \boldsymbol{\varphi} \end{bmatrix} + \begin{bmatrix} \mathbf{R} & \mathbf{A} \\ -\mathbf{A}^T & \mathbf{G} \end{bmatrix} \begin{bmatrix} \mathbf{i} \\ \boldsymbol{\varphi} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{i}_s \end{bmatrix}$$

Boundary conditions

- The electric potential is defined up to an additive constant
- The current is injected in the first layer and taken in the last



Boundary conditions

- One face can be set to zero potential

$$\varphi_1 = \varphi_2 = \varphi_3 = 0$$

potential FIRST layer
↗

- The other face potential are identical but to an unknown value

$$\varphi_7 = \varphi_8 = \varphi_9 = \varphi^*$$

All solution ARE
EQUAL

Fixed potential (Dirichlet)

- Suppose to have a linear system with some unknown at fixed value x_D (Dirichlet bc)

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_D \end{bmatrix} \begin{bmatrix} x_1 \\ x_D \end{bmatrix} = \begin{bmatrix} b_1 \\ b_D \end{bmatrix}$$

- Solving the first block row

known - move it to the right

$$A_{11}x_1 + A_{12}x_D = b_1 \Rightarrow A_{11}x_1 = b_1 - A_{12}x_D$$

- The second block row has no interest

Floating potential

- For simplicity consider a 3×3 system with $x_1 = x_3$ (unknown)

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

- The first row is

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_1 = b_1$$

$$(a_{11} + a_{13})x_1 + a_{12}x_2 = b_1$$

- Sum of cols

$$\begin{bmatrix} a_{11} + a_{13} & a_{12} \\ a_{21} + a_{23} & a_{22} \\ a_{31} + a_{33} & a_{32} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

3x2 we don't like it

- Sum of rows

$$\begin{bmatrix} a_{11} + a_{13} + a_{31} + a_{33} & a_{12} + a_{32} \\ a_{21} + a_{23} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 + b_3 \\ b_2 \end{bmatrix}$$

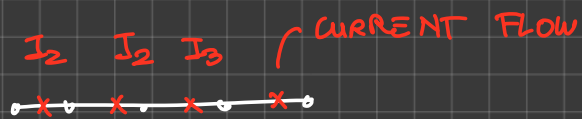
this way
symmetric preserve
matrix

— ANATOMY OF THE CODE



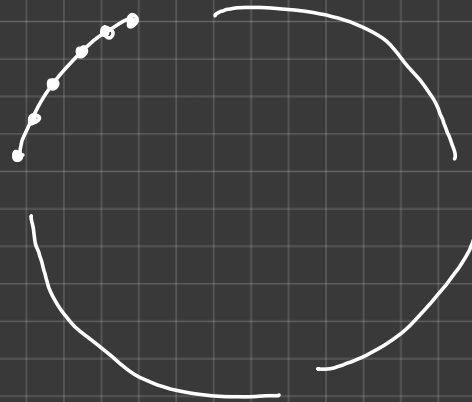
① — preprocess geometry in order to simplify the problem

let's go 2D



/ TAPE

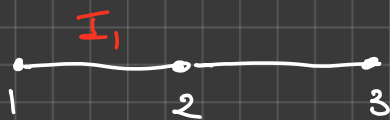
set of point to
identify the TAPE



TAPE AROUND CORE

ONLY DC WORKING CONDITION

→ Define mesh size | REMEMBER THAT they infinit. thin conductors



— MATERIAL PROPERTY — RESISTIVITY non linear

$$p(B_u, B_t, J, T) = \frac{E_c}{J} \left(\frac{J}{J_c(B_u, B_t, T)} \right)^n$$

$$J_c(B_u, B_t, T) = \frac{J_{co}}{1 + \left(\frac{\sqrt{K B_t^2 + B_u}}{B_0} \right)^\alpha} \frac{T_c - T}{T_c - T_{op}}$$

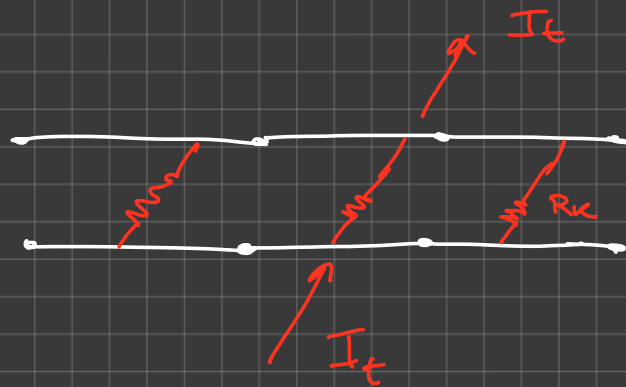
Given

operational

(2) solver — Assembly the MAT of final sys

How to assemble

They can be curved → linear polygonal



each can be rapp. aso resistance $R_k = \rho_k (B_n, B_e, J, T) \frac{L}{S_k}$ $S_k = n_k \delta_{\text{min}} \mu\text{m}$
 in the layer above

$$V_k = R_k I_k = \rho_k \frac{L}{S_k} I_k = \rho_k L J_k \quad \forall k$$

different value of current? **audio 1:14**

there only 1 element
 all resistance are in //

$V_1 = V_2 = \dots V_k = V_n = V$ BC same voltage

$$\sum_{k=1}^N I_k = I_t$$

$$\begin{bmatrix} R_1 & & & & \\ & R_2 & & & \\ & & \ddots & & \\ & & & R_k & \\ & & & & R_n \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_k \\ \vdots \\ I_n \end{bmatrix} = \begin{bmatrix} V \\ V \\ V \\ \vdots \\ V \\ I_t \end{bmatrix}$$

sum of total current have to be I_t

BUT the V are known — they must go on the left

$$\begin{bmatrix} R_1 & & & -1 \\ & R_2 & & -1 \\ & & \ddots & \vdots \\ & & & R_k & -1 \\ & & & & \vdots \\ 1 & 1 & \dots & 1 & \dots & 0 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_n \\ V \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ I_t \end{bmatrix}$$

the RES ARE FUNC OF η , BUT DEPENDS ON J AND B

singular
element

All currents

we CAN SAY

$A(x) x = b \rightarrow$ non linear sys

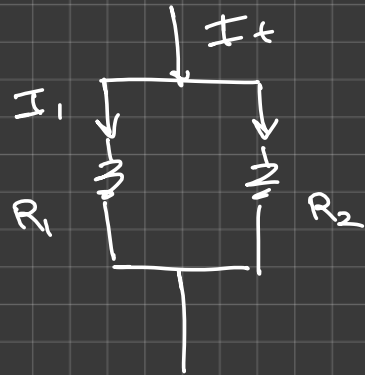
2.b Additional Functions : plot-SAVAR-Like Machinery

2.c Solution

Fsolve
(MATLAB)

$$F(x) = 0$$

MUST BE PROVIDED
↓
 $F(x) = A(x) x - b$



$$I_1 = \frac{R_2}{R_1 + R_2} I_t$$

$$I_1 = \frac{G_1}{G_1 + G_2} I_t$$

$$I_1 = \frac{\frac{1}{R_1}}{\frac{1}{R_1} + \frac{1}{R_2}} I_t$$



3 unk Fsolve $\rightarrow V^{(1)}$

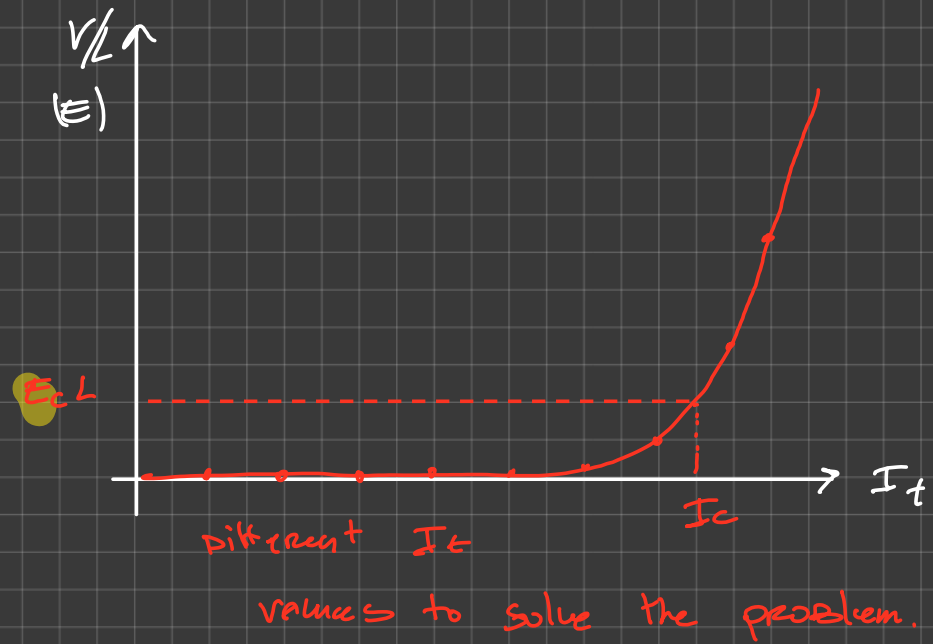
6 unk Fsolve $\rightarrow V^{(2)}$

$\hookrightarrow$ 6 current 1 voltage

two different refinement

error $\epsilon^{(2)} = \frac{V^{(2)} - V^{(1)}}{V^{(1)}}$

ESTIMATION of how good is the solution



length independent graph



Audio 1:58

We get the VI CHARACTERISTIC

